

Datacenter load, demand response, and resource adequacy

PJM and ERCOT · ReEDS 2032 · PRAS

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Datacenter load, demand response, and resource adequacy — brief results

PJM and ERCOT, ReEDS model year 2032 · PRAS · 15 weather years (2007-2021) × 1000 Monte Carlo samples Prepared 2026-07-22 · reproducible via scripts/00-04, see [README.md](#)

What was done

ReEDS built a 2032 grid under the **default** demand forecast — every generator, battery and transmission line. That fleet was then **frozen**.

PRAS took the frozen fleet and simulated the year hour by hour, 8760 hours × 15 weather years, 1000 times over, randomly failing generators at their forced-outage rates. Each hour it asks whether generation plus imports could serve load; when they could not, it records the shortfall. **NEUE** (normalized expected unserved energy, in parts per million) is that shortfall as a share of total load.

That simulation was run twice — once with normal load, once with datacenter load added on top — **using the same power plants both times**. Then demand response was layered on: each region received a DR device sized as a fraction of *its own added datacenter load*, swept from 5 % to 100 %, in two flavors:

- **shift** — load moved to a later hour and repaid within the payback window; available 24/7.
- **shed** — load dropped and never repaid; available only in a fixed daily window.

Because the buildout never changes, any difference in NEUE is attributable to the added load and to the DR — not to different generation. The question being answered is deliberately narrow: **can flexible datacenter load substitute for generation that was never built?**

92 PRAS runs in total: 68 PJM, 24 ERCOT.

Headline results

1. ERCOT shows no reliability impact at all

All 24 ERCOT runs report **exactly zero** unserved energy — both baselines and every DR case. This is not a failed run: the added load is present (+8 879 MW at peak), but ERCOT enters 2032 with ~41 % reserve margin at peak against PJM's 28 %, and a mean forced-outage rate about a third of PJM's. It simply never runs short.

Everything below is therefore PJM only.

2. Datacenter load raises PJM's NEUE 21x

case	NEUE (ppm)	EUE (MWh over 15 weather years)
base demand, no DR	0.0086	113
+ datacenter load, no DR	0.1796	2 565
+ datacenter load, best DR (shift_16h, 100 %)	0.0932	1 332

Even the best DR case leaves NEUE ~11x above the base-demand level. **DR meaningfully mitigates but does not neutralize the added load.**

For context, PJM's reliability standard is conventionally 1 day in 10 years; all of these NEUE values are small in absolute terms. The story here is the *relative* change and *where* it lands.

3. Nearly all of it is one small region

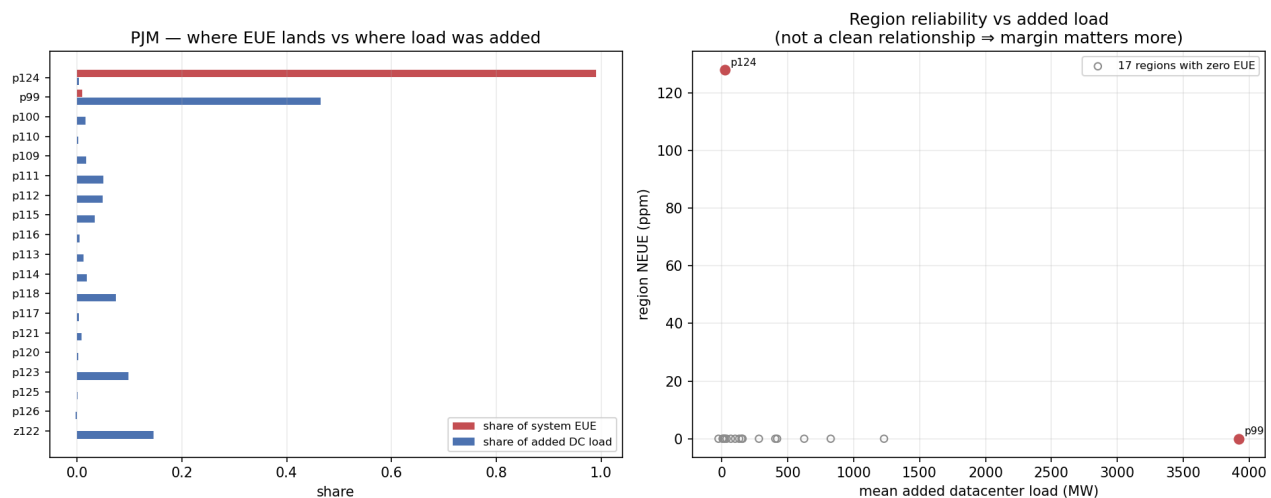


Figure 1: where

99.0 % of PJM's unserved energy occurs in region p124, which receives only **0.3 % of the added datacenter load**. Meanwhile p99 absorbs 47 % of the added load and contributes 0.9 % of the EUE.

p124 is a small, wind-dependent, weakly-connected pocket:

p124	value
peak load	220 MW
firm (non-VRE) capacity	60 MW
wind capacity	up to 2 611 MW
storage	148 MW
import capability	80 MW (lowest ratio to peak load in PJM)
hours with negative local margin	11.2 %
hours negative <i>even with full imports</i>	2.0 %

So p124 covers a 220 MW peak with 60 MW of firm capacity, leaning on wind it cannot replace and cannot import around. Adding ~30 MW of datacenter load — a 14 % increase on its peak — pushes it over. **This is a local deliverability problem that datacenter load exposed, not a system-wide capacity shortfall.**

Consistent with that: only **1 % of the 248 outage episodes involve more than one region.** These are isolated local events, not correlated system-wide scarcity.

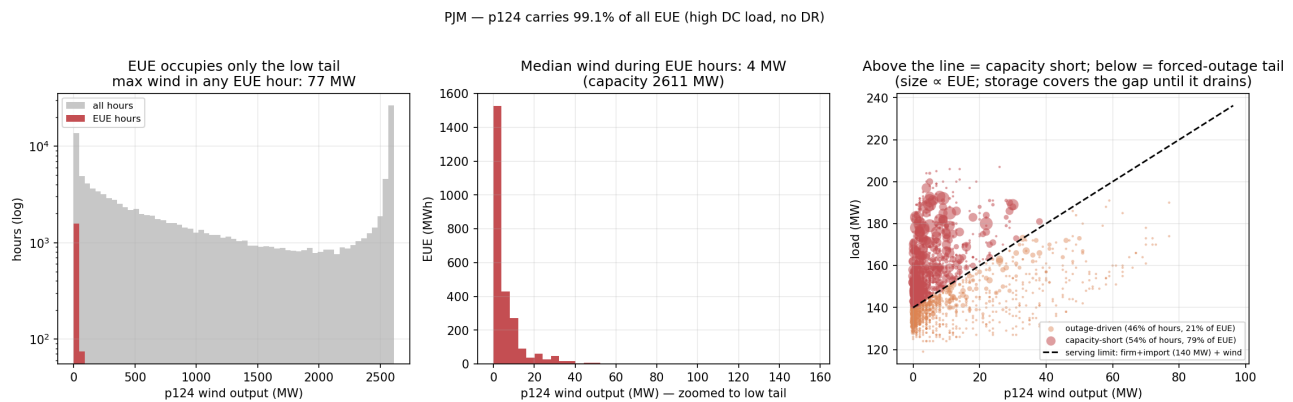


Figure 2: critical region

p124 fails only in the low tail of its own wind output 100 % of p124's EUE occurs when its wind output is ≤ 77 MW, against 2 611 MW of capacity and an all-hours median of 1 009 MW. Median wind during EUE hours is **4 MW**. Only 14 % of EUE hours have wind at exactly zero, but 97 % are at or below 50 MW and those carry 99.8 % of the energy — so the driver is a wind drought, not literal zero.

The arithmetic closes exactly:

	MW
firm capacity (constant)	60
import limit	80
servable without wind	140
load during EUE hours	median 150, max 207

p124 is short by ~10 MW at median load and ~67 MW at peak, so it needs roughly **70 MW of wind — 2.7 % of its installed capacity** — to close the gap, or the battery until it drains at 2.9 h. That threshold is why no EUE hour ever has more than 77 MW of wind. Given wind at exactly zero (790 h over 15 weather years), 29.5 % of those hours produce EUE; the rest are covered by storage or low load.

This makes the stranded-wind observation a red herring for adequacy. The 2.5 GW is irrelevant to reliability except through the bottom ~3 % of its output distribution. It also explains why a 30 MW DR device covers so much of the gap: the deficit is genuinely shallow.

Two distinct failure modes appear, and the split matters:

	hours	share of EUE	mean EUE/hour
capacity-short (load > firm + import + VRE)	628 (38 %)	72 %	2.90 MWh
outage-driven (nameplate adequate; forced-outage tail)	1 013 (62 %)	28 %	0.71 MWh

Because /eue is a *mean over 1000 Monte Carlo samples*, an hour can carry a small positive expectation even when nameplate capacity covers load — a minority of draws lose part of the 60 MW firm fleet or the tie. Most hours are of this type, but they contribute little energy; the real damage is the 628 genuinely-short hours.

Structural detail

p124 structure	
transmission	a single radial tie to p123: 80 MW in, 336 MW out
offshore wind	2 536 MW — 11.5× the zone’s own peak load
onshore wind / solar	75 MW / 97 MW
firm generation	22 oil-gas-steam units totalling 60 MW ; twelve are 0 MW, most of the rest are 1–2 MW
battery	426 MWh / 148 MW = 2.9 h

The arithmetic fails in both directions:

- **Outbound:** 167 TWh of VRE generated against 19.9 TWh of local load, behind 336 MW of export capacity. **116 TWh (69.6 %) is stranded** — in 66 % of all hours p124 produces more than it can consume or ship out.
- **Inbound:** 11.2 % of hours have load above VRE + firm; after the 80 MW import limit, 2 603 hours (2.0 %) remain short. Those deficits are **shallow but long** — a 55 MW maximum, but **386 of 853 episodes outrun the battery’s 2.9 h**.

That last line is the mechanism behind every PJM result in this study: a wind drought lasting more than about three hours, in a zone with 60 MW of real generation behind an 80 MW tie. It also explains why a mere 30 MW DR device is so effective there — the deficit never exceeds 55 MW.

This is not a translation bug. Two tempting explanations were checked and both fail:

- “*The offshore wind’s interconnection is missing.*” No — the plant sits inside p124, so no inter-connection to it is required, and p124 exports through the 336 MW backward capacity on its tie. Confirmed against the [PRAS HDF5 spec](#): forwardcapacity is region_from→region_to, so the p123→p124 line gives p124 80 MW in and 336 MW out.
- “*The 80/336 asymmetry is anomalous.*” No — **all 41 PJM lines are asymmetric, none symmetric**. Directional ratings are simply how ReEDS transmission translates into PRAS. p124’s 4.2× ratio is unremarkable next to p110/p118’s 13×.

So p124 is genuinely capacity-deficient **in the modeled system**, and that is a real property of what ReEDS built rather than an artifact of the PRAS conversion.

The likely explanation is policy, not error. The ReEDS scenario is `high_currentpolicy_central`, and current policy includes **state offshore-wind mandates**. ReEDS would be obliged to build that 2 536 MW regardless of deliverability; if it did not co-optimize a matching tie, the result is exactly what is observed — policy-driven offshore wind, ~70 % of it curtailed, in a balancing area with 60 MW of firm capacity. Ordinary model behavior under a binding constraint.

What this means for interpretation: the result is real within the model, but narrow. It says a small coastal BA meeting an offshore-wind mandate has thin firm capacity behind a small tie, and ~30 MW of datacenter load tips it over. It does **not** say PJM at large has a datacenter-driven adequacy problem.

Worth confirming: does the ReEDS run itself show comparable curtailment in p124? If so, this is settled as expected behavior rather than a data issue.

4. The failures are low-wind, high-net-load evening hours

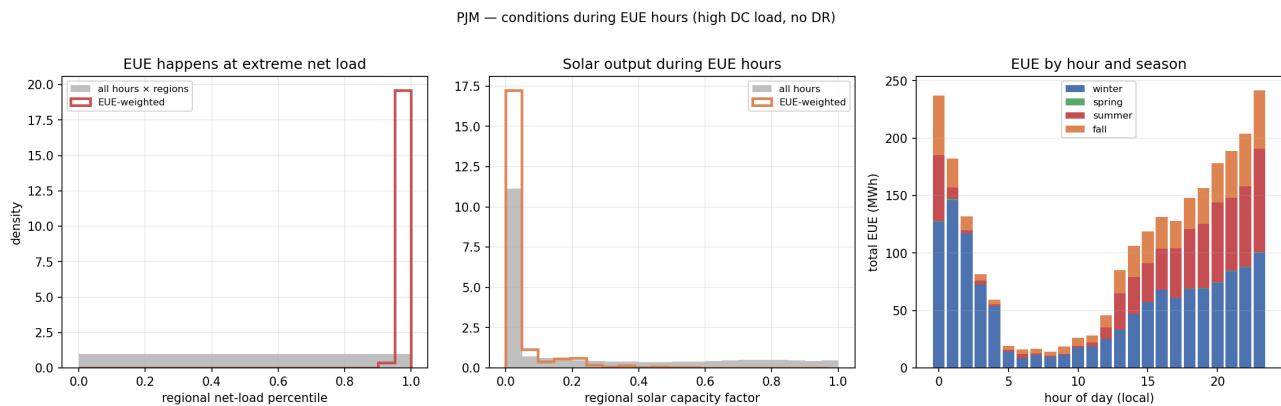


Figure 3: conditions

condition	all hours	during EUE (EUE-weighted)
regional net-load percentile	0.50	0.986
regional margin percentile	0.50	0.014
wind capacity factor	0.425	0.005
solar capacity factor	0.227	0.032

The wind number is the mechanism. In a region that depends on wind for capacity, EUE occurs when wind output is at **half a percent** of its maximum. Events cluster in evening and overnight hours (peaking 8 pm–1 am local), split between winter and summer, with essentially nothing in spring.

5. Adding datacenter load moves the problem, not just its size

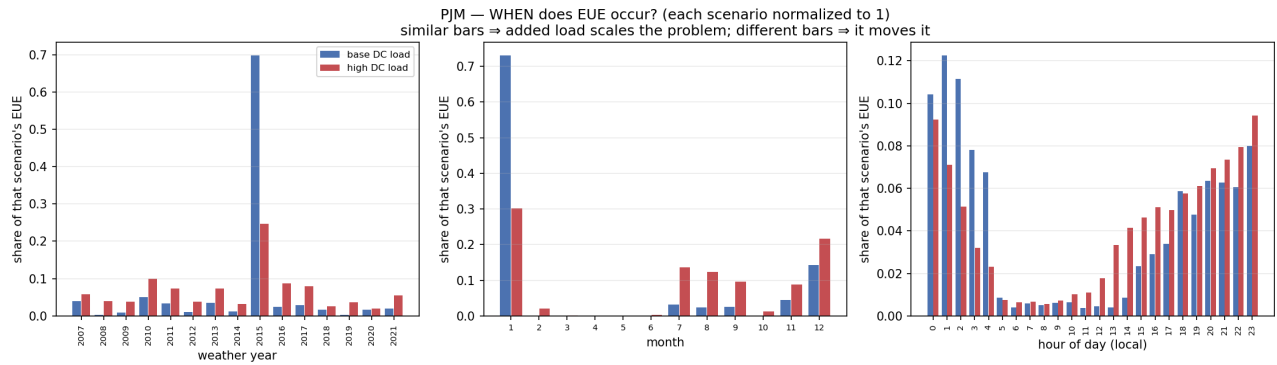


Figure 4: when

Comparing *normalized* distributions between the base and high scenarios (total variation distance; 0 = identical shape):

dimension	shift
weather year	0.45
month	0.43
hour of day	0.22

Weather-year and seasonal patterns change substantially — **41 % of the high case's EUE occurs in hours that had none at all in the base case**. So the added load is not simply scaling up existing bad hours; it is creating new failure hours in different weather years and seasons. Hour-of-day shifts much less, which fits a local capacity deficit rather than a change in daily shape.

6. DR helps, gradually, with no useful cut-off point

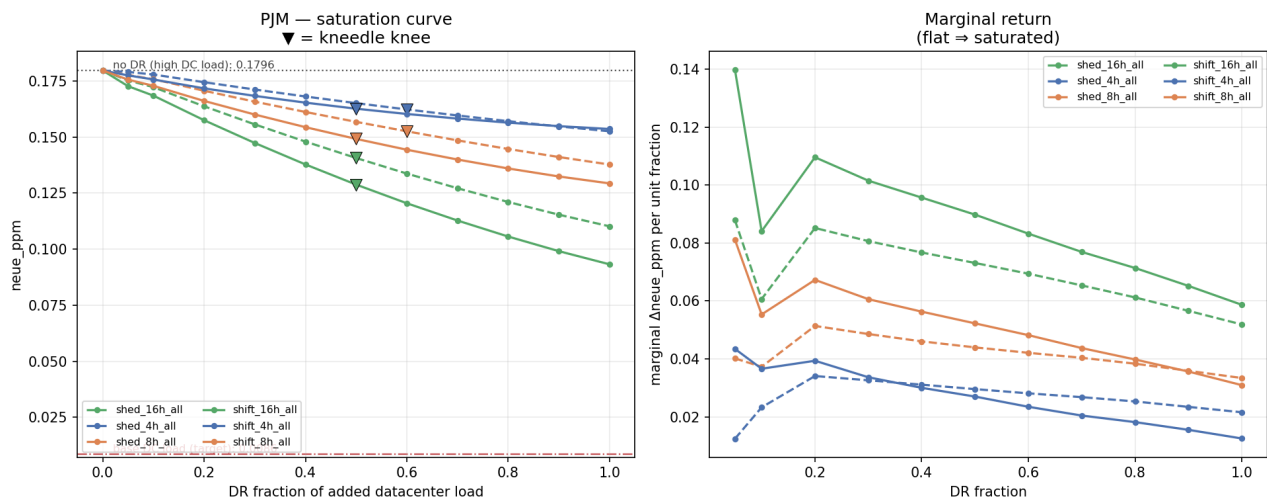


Figure 5: saturation

DR family	NEUE floor at 100 %	share of datacenter penalty recovered
shift 16 h	0.0932	50.5 %
shed 16 h	0.1102	40.6 %
shift 8 h	0.1293	29.4 %
shed 8 h	0.1377	24.5 %
shed 4 h	0.1525	15.8 %
shift 4 h	0.1536	15.2 %

There is **no sharp knee** — no fraction past which added DR stops helping. Returns decline smoothly; reaching 80 % of everything achievable needs a DR fraction of roughly 0.7–0.8.

Device duration matters far more than deployment share. Going 4 h → 16 h roughly triples the benefit; raising the fraction within a family does much less.

Most of that benefit comes from 30 MW in the right place. In the best case, p124's 30 MW DR device delivers 1 210 MWh of the total 1 234 MWh reduction. The 4 696 MW of DR sitting in p99 contributes almost nothing, because p99 has almost no EUE to fix. Sizing DR to where datacenter load *is* turns out to be nearly orthogonal to where reliability risk *is*.

7. The shed-vs-shift comparison in this sweep is not valid

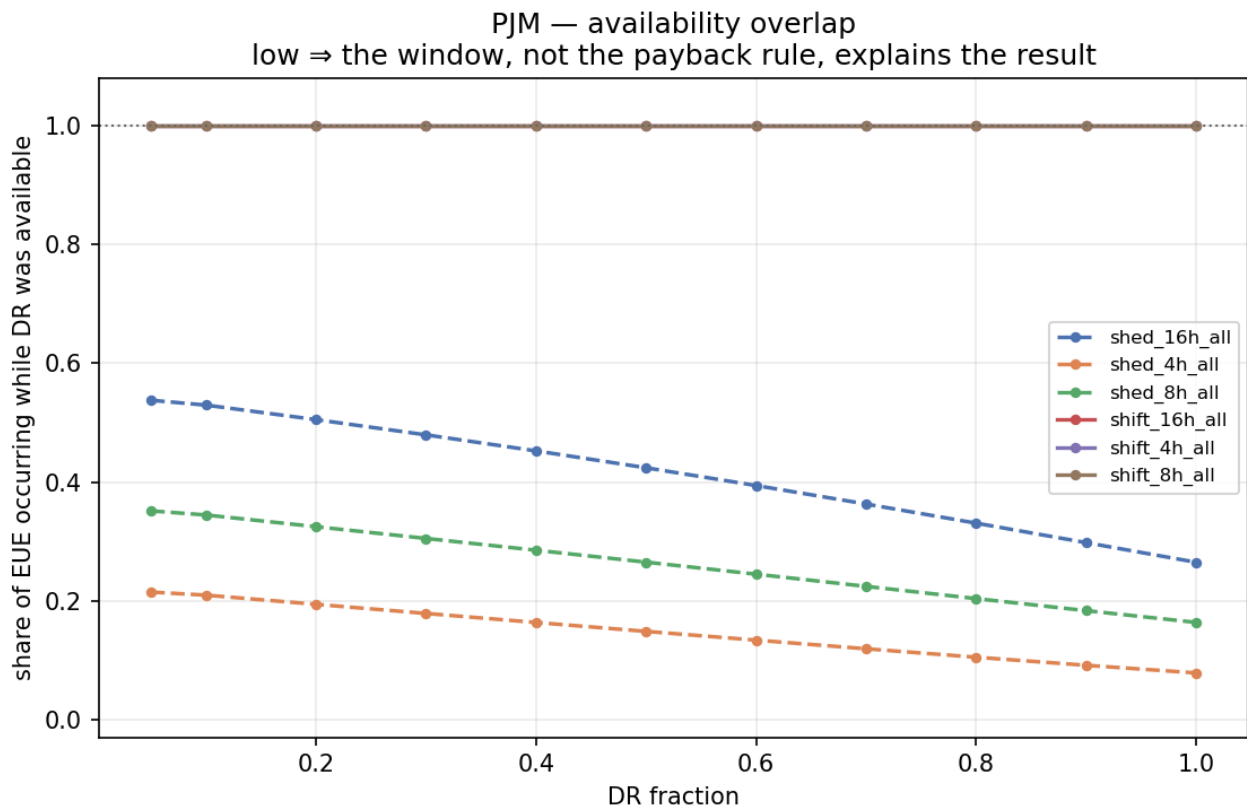


Figure 6: overlap

Shed and shift differ in **two** ways at once — shed forgives the payback (an advantage) but is restricted to a daily window (a handicap). They push in opposite directions, so the comparison

cannot be read as-is.

Measuring the share of each case's remaining EUE that falls inside its own DR window settles it: at 100 % deployment, only **8 %** of shed_4h's EUE occurs when it is allowed to run. Shed is losing on availability, not on mechanism. The overlap *declines* as deployment rises — DR removes the EUE it can reach, so what survives is increasingly outside the window.

An always-on shed run is needed to compare the mechanisms. That is a config change (drop available_hours_et), not new code.

What this means

1. **The binding constraint is local, not systemic.** PJM's added datacenter load did not stress the system as a whole; it exposed one small wind-dependent region with 60 MW of firm capacity and 80 MW of import capability. Whether p124 is a genuine reliability concern or an artifact of regional aggregation in ReEDS/reeds2pras **should be checked before this result is used** — a 220 MW region holding 2.6 GW of wind is unusual.
 2. **DR sized to datacenter load is poorly targeted.** The regions receiving the datacenter load are largely not the regions with reliability risk. A DR program sized to local *adequacy need* rather than to local *datacenter megawatts* would do far more per MW.
 3. **Duration beats quantity.** If there is a design lever in these results, it is longer-duration flexibility, not more of it.
 4. **ERCOT's null result is itself informative** — that buildout absorbs central-case datacenter growth without measurable reliability cost.
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Caveats

- **Monte Carlo noise.** Case-level standard error is ~1 % of EUE, so NEUE stderr is about ± 0.0018 — *larger* than most step-to-step differences along the DR sweep. The curves are smooth because all cases share a random seed (common random numbers), making paired differences more reliable than the individual error bars imply. This cannot be quantified from the saved outputs, which do not retain per-sample draws. **Treat single-increment differences as indicative, not significant.**
 - **One region drives everything.** With 99 % of EUE in p124, every aggregate PJM statistic here is effectively a statement about p124. Conclusions are only as robust as that region's representation.
 - **Same buildout by construction.** Generation was not re-optimized for the higher load. These results describe what happens if datacenter load arrives *without* a supply response — not what a planner would actually build.
 - **Event counts depend on a threshold.** Episode counts range from 294 to 7 as the EUE floor moves from 0.01 to 10 MWh (pjm_eps_sensitivity.csv). Totals are robust; counts are not.
 - **The DR device is regional, not datacenter-specific.** It is *sized* from datacenter load, but PRAS reduces total regional load when it operates — it does not track whose megawatt-hour is whose.
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Files

Output	Path
Figures	outputs/figures/pjm_{where,when,conditions,worst_e
Regional attribution	outputs/tables/pjm_where.csv
Timing	outputs/tables/pjm_when.csv, pjm_shape_shift.csv
Event conditions	outputs/tables/pjm_why.csv
New vs amplified failures	outputs/tables/pjm_delta.csv
Episodes	outputs/tables/pjm_events.csv, pjm_concentration.csv, pjm_eps_sensitivity.csv
DR saturation	outputs/tables/dr_saturation_{curves,summary}.csv, dr_limit_diagnosis.csv
Case registry	cache/registry.csv

Full detail, including data-quality issues found in the source files, is in [FINDINGS.md](#).